



Module 1: Networking Today

Telecom Academy

Introduction to Networks v7.0
(ITN)



Instructor-led, Online self-paced

Get Connected

Learn basic computer skills such as social media, Internet navigation & simple networking.



Beginning



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NDG Linux Unhatched

The "start from scratch" Linux course. A quick intro to the popular operating system.



Beginning



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IT Essentials

Learn computer fundamentals and validate skills through optional certification.



Intermediate



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NDG Linux Essentials

Build upon your existing Linux skills and prepare for certification.



Intermediate



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NDG Linux I

The first in a 2-course series. Begin your Linux mastery and prepare for certification.



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The second in a 2-course series. Further your mastery of Linux and prepare for certification.



Advanced



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Networking Essentials

Learn basic networking concepts and skills you can put to use right away.



Intermediate



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Instructor-led

CCNA: Introduction to Networks

The first course in a 3-course CCNA series designed to prepare you for entry-level networking job



Intermediate



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CCNA: Switching, Routing, and Wireless Essentials

The second course in a 3-course CCNA series designed to prepare you for networking technician job



Intermediate



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CCNA: Enterprise Networking, Security, and Automation

The third course in a 3-course CCNA series prepares you for networking analyst & engineering



Advanced



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CCNP Enterprise: Core Networking

The first course in a 2-part series prepares you for professional-level networking roles.



Advanced



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Instructor-led

CCNP Enterprise: Advanced Routing

The second course in a 2-course CCNP series prepares you for professional-level networking roles.



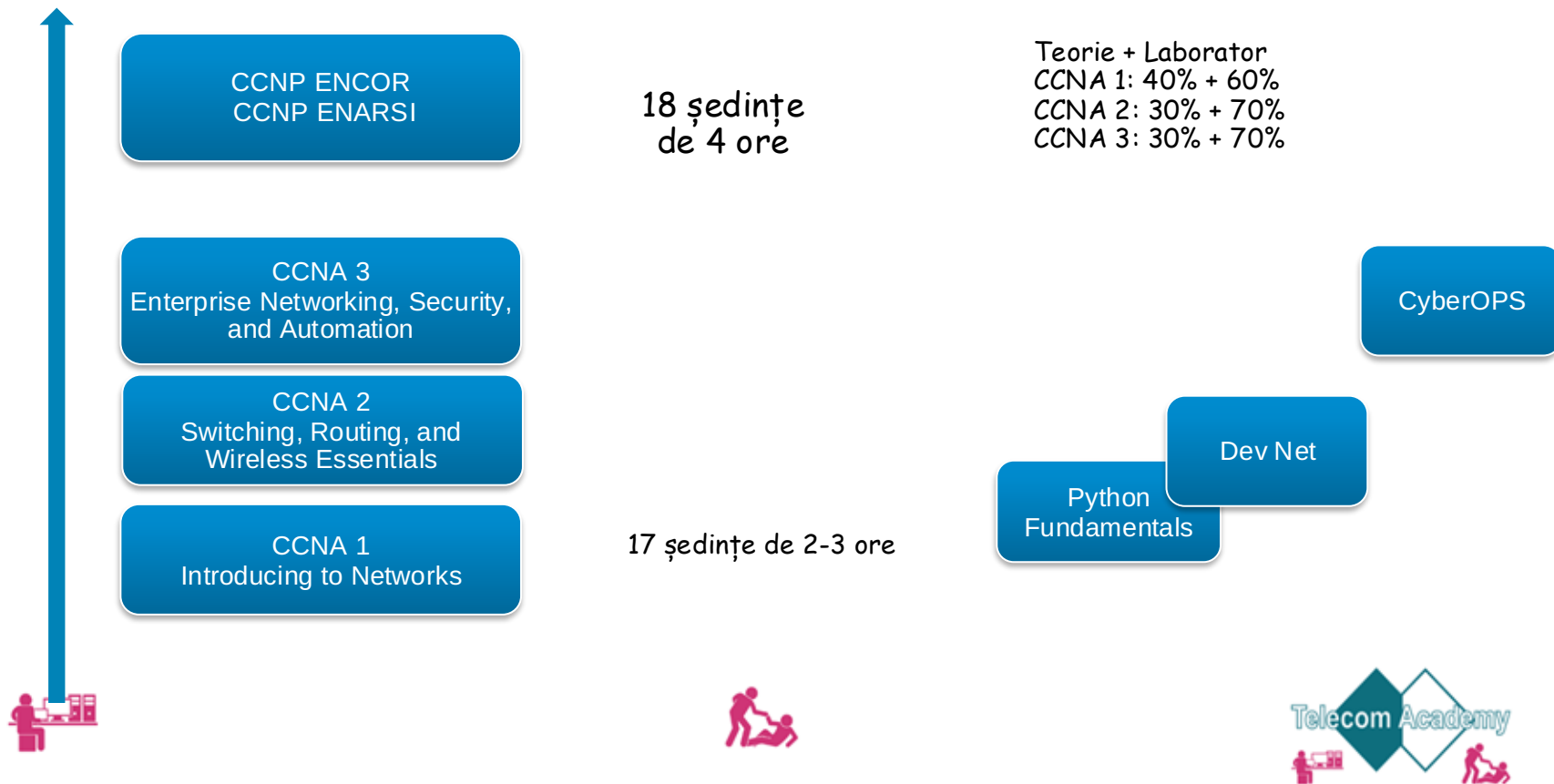
Advanced



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Mod de desfășurare - Curs CCNA 1



Diplomă



Certificate of Course Completion

Cisco Networking Academy

CCNAv7: Introduction to Networks

The student has successfully achieved student level credential for completing CCNAv7: Introduction to Networks course administered by the undersigned instructor. The student was able to proficiently:

- Configure switches and end devices to provide access to local and remote network resources.
- Explain how physical and data link layer protocols support the operation of Ethernet in a switched network.
- Configure routers to enable end-to-end connectivity between remote devices.
- Create IPv4 and IPv6 addressing schemes and verify network connectivity between devices
- Explain how the upper layers of the OSI model support network applications.
- Configure a small network with security best practices.
- Troubleshoot connectivity in a small network.

Viad

Student

Telecom Academy

Academy Name

Romania

Location

9 Dec 2022

Date

Cristian Deacu

Instructor

Instructor Signature



Certificarea CCNA



Cisco Certifications

Amer Jaber

has successfully completed the Cisco certification exam requirements and is recognized as a

Cisco Certified Network Associate Routing and Switching



Date Certified September 16, 2015
Valid Through September 16, 2018
Cisco ID No. CSC012874693

A handwritten signature in black ink, reading "Chuck Robbins".

Chuck Robbins
Chief Executive Officer
Cisco Systems, Inc.

Validate this certificate's authenticity at
www.cisco.com/go/verifycertificate
Certificate Verification No. 424135227652CLVN

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Calendar

CCNA 1 - 1 Martie 2023

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Congratulations! You have completed your Cisco Networking Academy course!

Discount Voucher

A discount voucher for the certification exam is available after you have completed the **full CCNA series of courses**.

If you scored a 70% or higher on the **FIRST** attempt of the **Final Exam in the last course, ENSA**, and have a "Pass" in the gradebook, then you are eligible for a discount voucher for a Cisco certification exam.

When you qualify for a Certification Exam discount (voucher) a link will display on your class tile in NetAcad.com and on your NetAcad.com profile.

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Access additional study materials or attend free webinars from Learning@Cisco

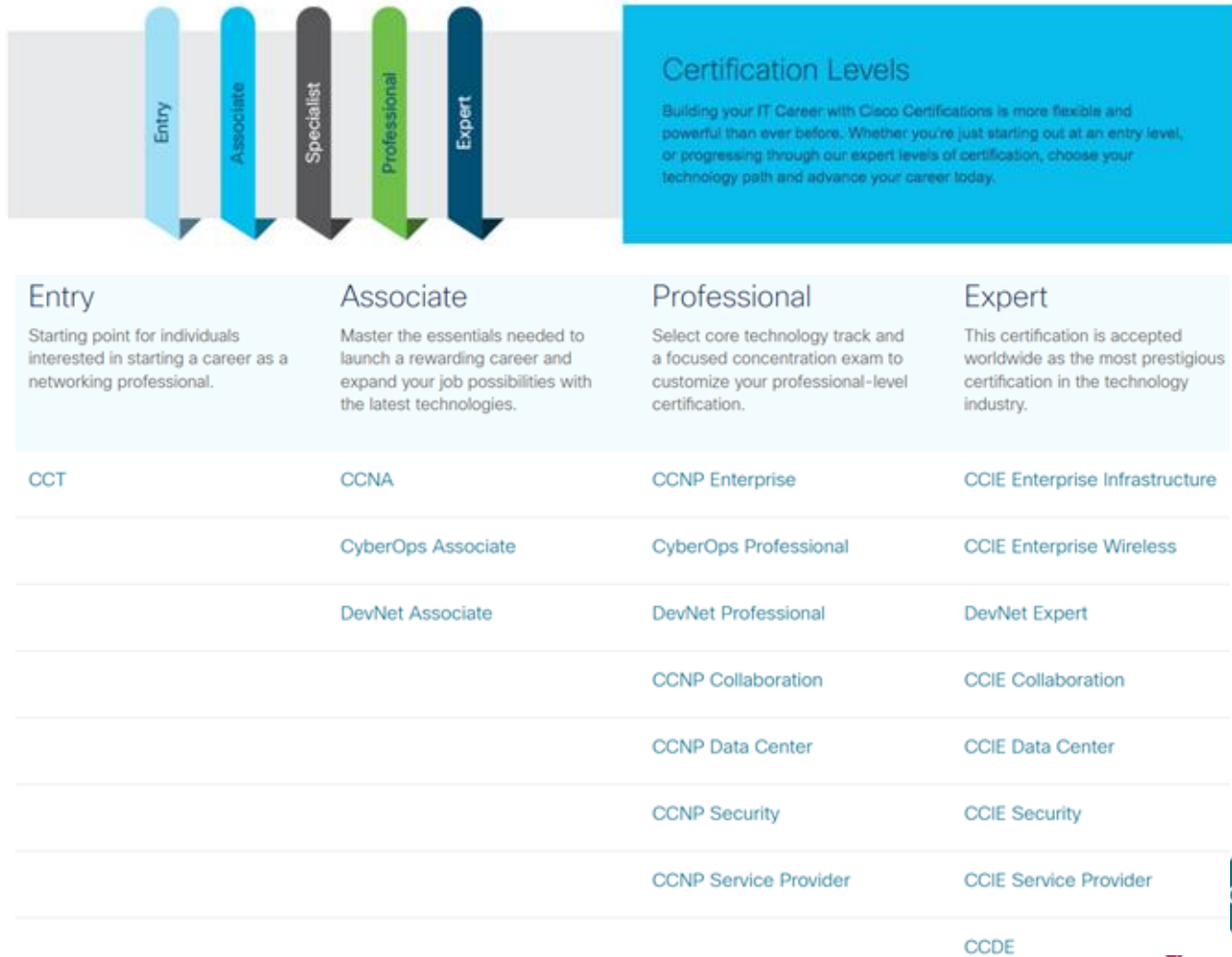
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Cisco Certifications



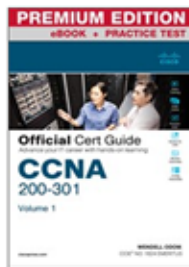


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CCIE Enterprise Infrastructure
Foundation, 2nd Edition



1 Basic Device Configuration

1.0 Introduction

1.0.1 Why should I take this module?

1.0.2 What will I learn to do in this module?

1.0.3 Video - Download and Install Packet Tracer

1.0.4 Video - Getting Started in Cisco Packet Tracer

1.0.5 Packet Tracer - Logical and Physical Mode Exploration

1.1 Configure a Switch with Initial Settings

1.2 Configure Switch Ports

1.3 Secure Remote Access

1.4 Basic Router Configuration

1.5 Verify Directly Connected Networks

1.6 Module Practice and Quiz

2 Switching Concepts

3 VLANs

Introduction

1.0.1

Why should I take this module?



Welcome to Basic Device Configuration!

Welcome to the first module in CCNA Switching, Routing, and Wireless Essentials! You know that switches and routers come with some built-in configuration, so why would you need to learn to further configure switches and routers?

Imagine that you purchased a model train set. After you had set it up, you realized that the track was just a simple oval shape and that the train cars only ran clockwise. You might want the track to be a figure eight shape with an overpass. You might want to have two trains that operate independently of each other and are able to move in different directions. How could you make that happen? You would need to reconfigure the track and the controls. It is the same with network devices. As a network administrator you need detailed control of the devices in your network. This means precisely configuring switches and routers so that your network does what you want it to do. This module has many Syntax Checker and Packet Tracer activities to help you develop these skills. Let's get started!

1.0.2

What will I learn to do in this module?



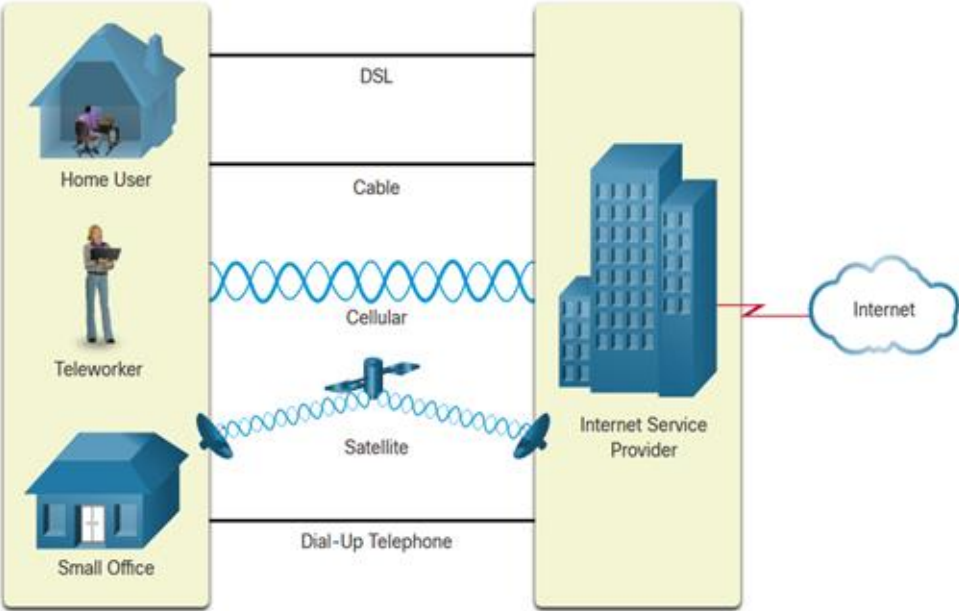
Module Title: Basic Device Configuration

Module Objective: Configure devices using security best practices.



Istoria Internetului

Home and Small Office Internet Connections



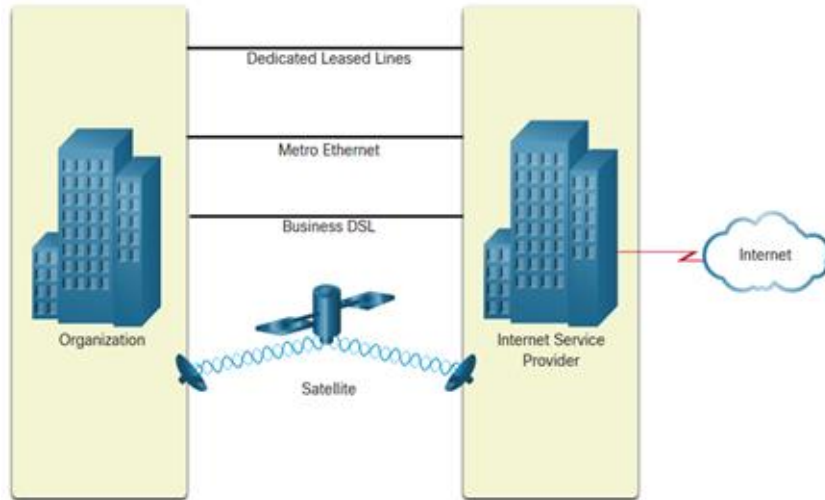
Connection	Description
Cable	high bandwidth, always on, internet offered by cable television service providers.
DSL	high bandwidth, always on, internet connection that runs over a telephone line.
Cellular	uses a cell phone network to connect to the internet.
Satellite	major benefit to rural areas without Internet Service Providers.
Dial-up telephone	an inexpensive, low bandwidth option using a modem.



Businesses Internet Connections

Corporate business connections may require:

- higher bandwidth
- dedicated connections
- managed services



Type of Connection	Description
Dedicated Leased Line	These are reserved circuits within the service provider's network that connect distant offices with private voice and/or data networking.
Ethernet WAN	This extends LAN access technology into the WAN.
DSL	Business DSL is available in various formats including Symmetric Digital Subscriber Lines (SDSL).
Satellite	This can provide a connection when a wired solution is not available.

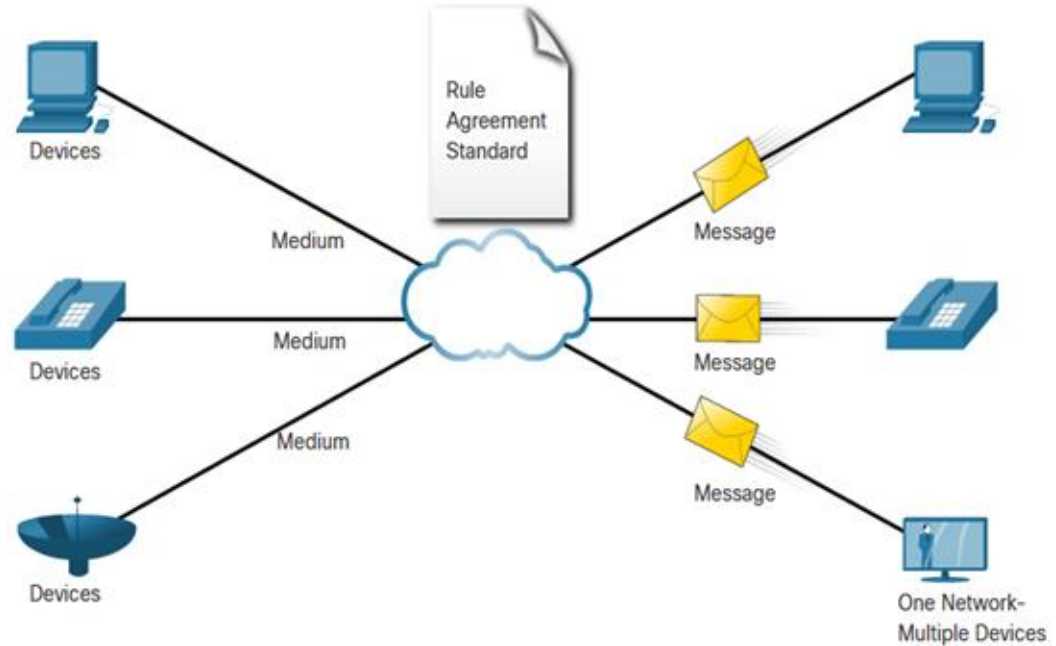


The Converging Network (Cont.)

Converged data networks carry multiple services on one link including:

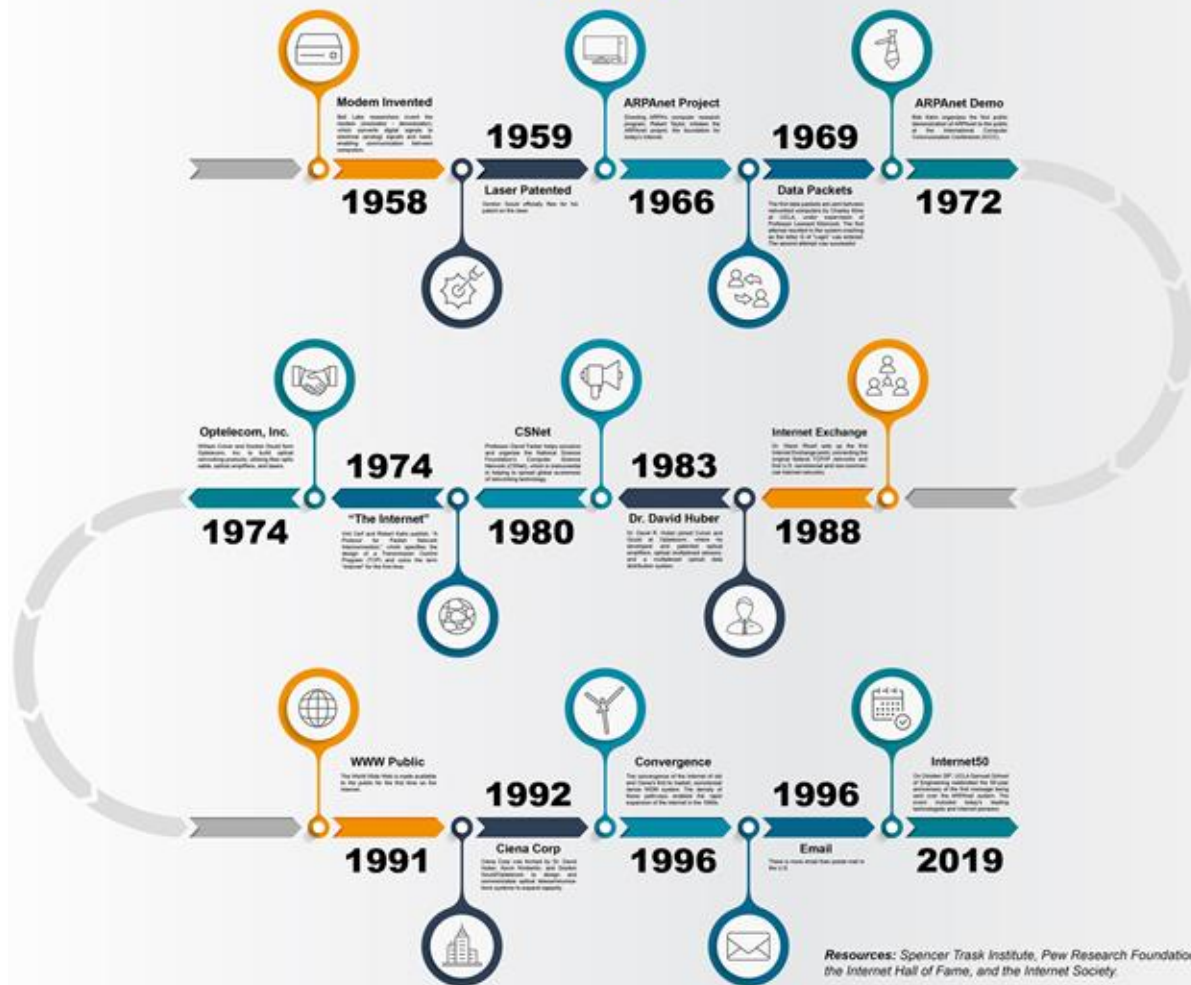
- data
- voice
- video

Converged networks can deliver data, voice, and video over the same network infrastructure. The network infrastructure uses the same set of rules and standards.



INTERNET HISTORY

Timeline



Resources: Spencer Trask Institute, Pew Research Foundation, the Internet Hall of Fame, and the Internet Society.



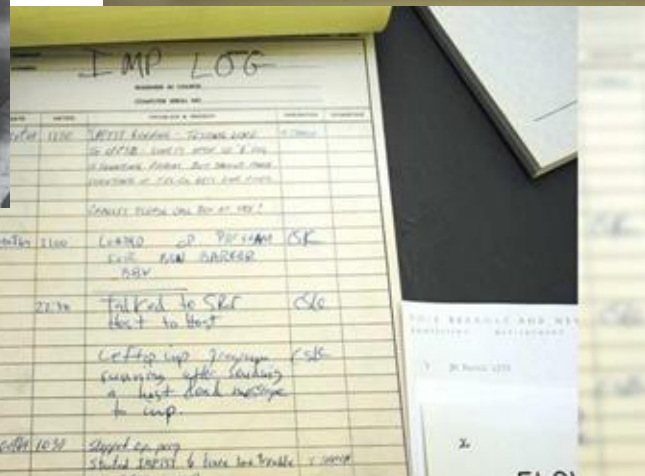
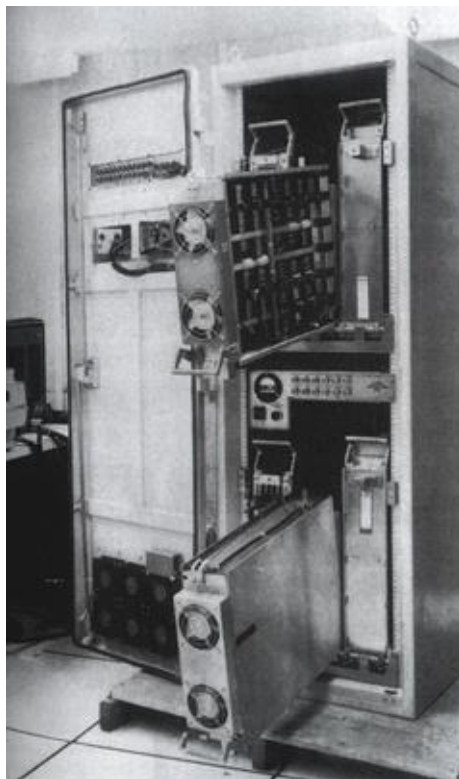




Arpanet

- The first Internet prototype created in 1965 by the Department of Defense.







7



8



9



Report No. 1837

Bolt Beranek and Newman Inc.

When all interrupts have finally been serviced, control is again returned to the background routine, which resumes at the instruction that was about to be executed when the first interrupt occurred.

B. Buffer Allocation and Queues

A large fraction of core memory is partitioned into fixed size buffers, each of which is used for storing a single packet. A buffer in which no packet is stored is said to be free, and all free buffers are chained together on a free buffer list. A buffer is removed from the free buffer list and set up for input each time a buffer is needed to store the next packet from a Host or Modem channel or to store an internally generated packet. Buffers that are not located on the free buffer list either are set up to store an incoming packet, are having their contents transmitted on one of the output channels, are being processed by the IMP program, or have been dispatched to one or more system queues for subsequent processing.

There are seven principal system queues:

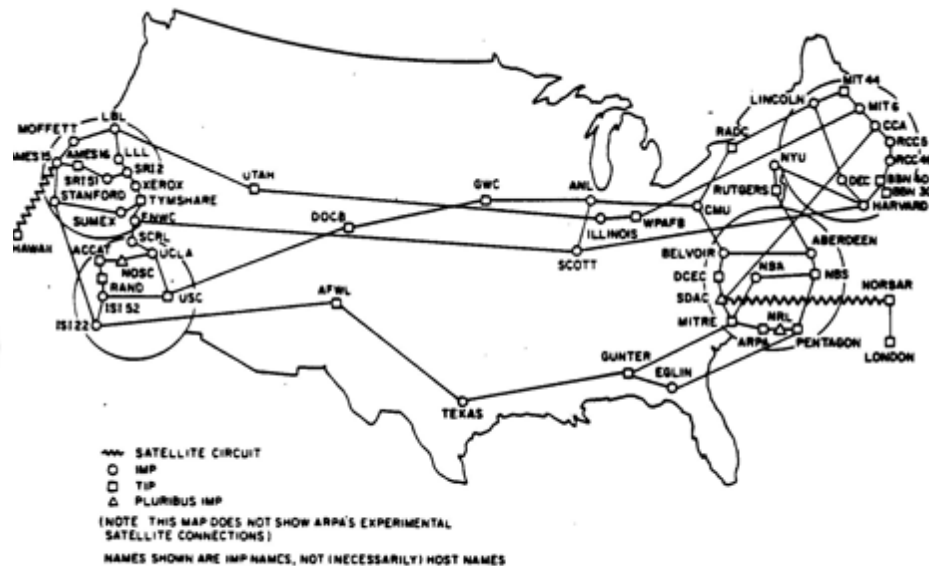
Task Queue

The task queue is organized into a Host task queue and two Modem task queues. Packets received on Host channels are placed on the Host task queue. All received acknowledgements, dead Host and routing information, I Heard You and Hello packets are placed on a fast Modem task queue. All other packets from the Modems



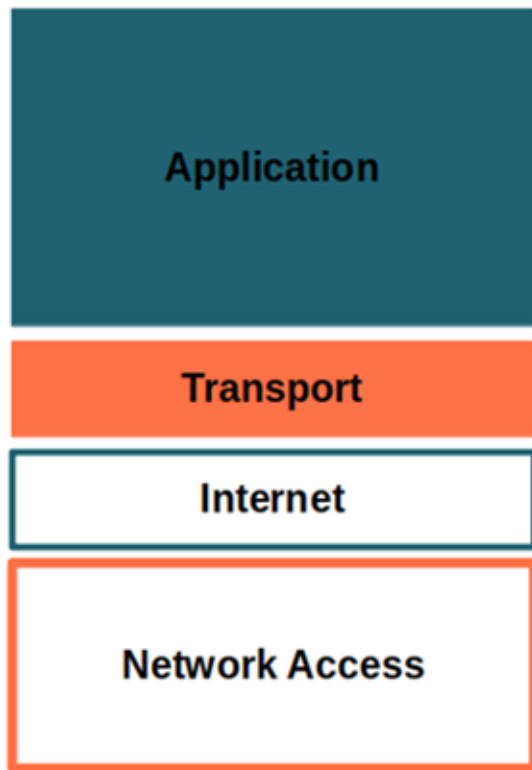


MAP 4 September 1971

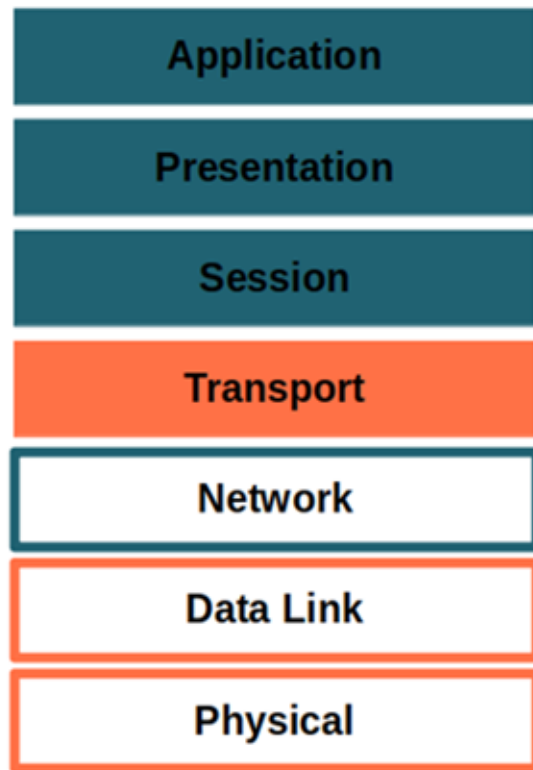


The TCP/IP Model vs. the OSI Model

The TCP/IP Model



The OSI Model



7 Layers of the OSI Model

Application

- End User layer
- HTTP, FTP, IRC, SSH, DNS

Presentation

- Syntax layer
- SSL, SSH, IMAP, FTP, MPEG, JPEG

Session

- Synch & send to port
- API's, Sockets, WinSock

Transport

- End-to-end connections
- TCP, UDP

Network

- Packets
- IP, ICMP, IPSec, IGMP

Data Link

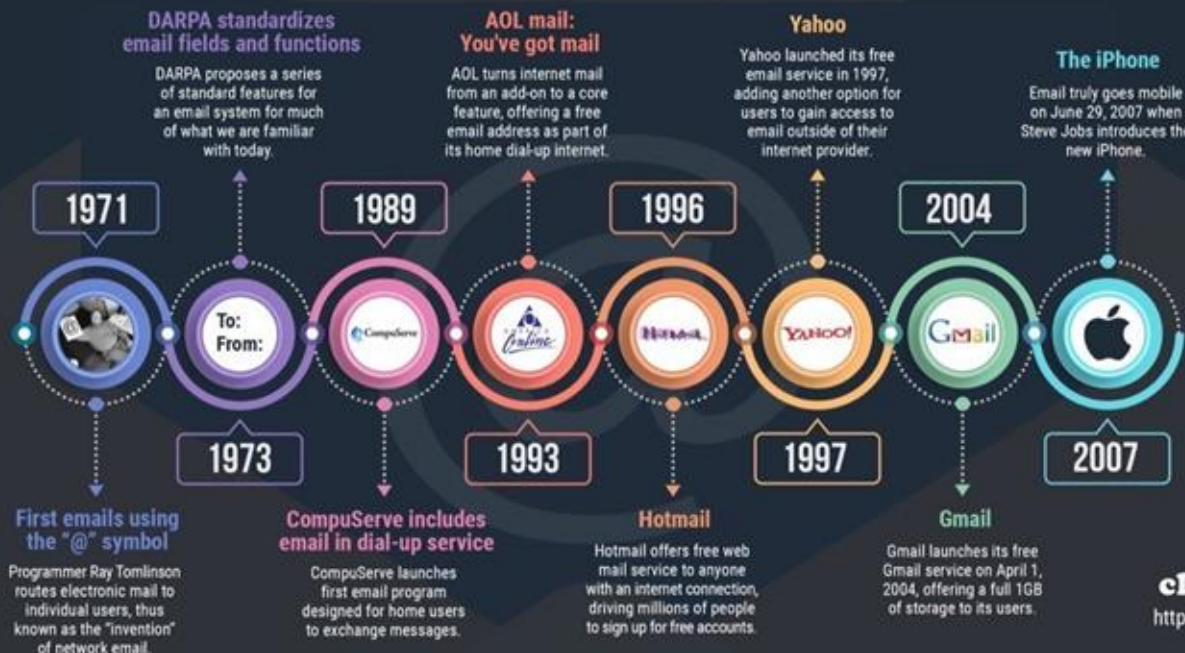
- Frames
- Ethernet, PPP, Switch, Bridge

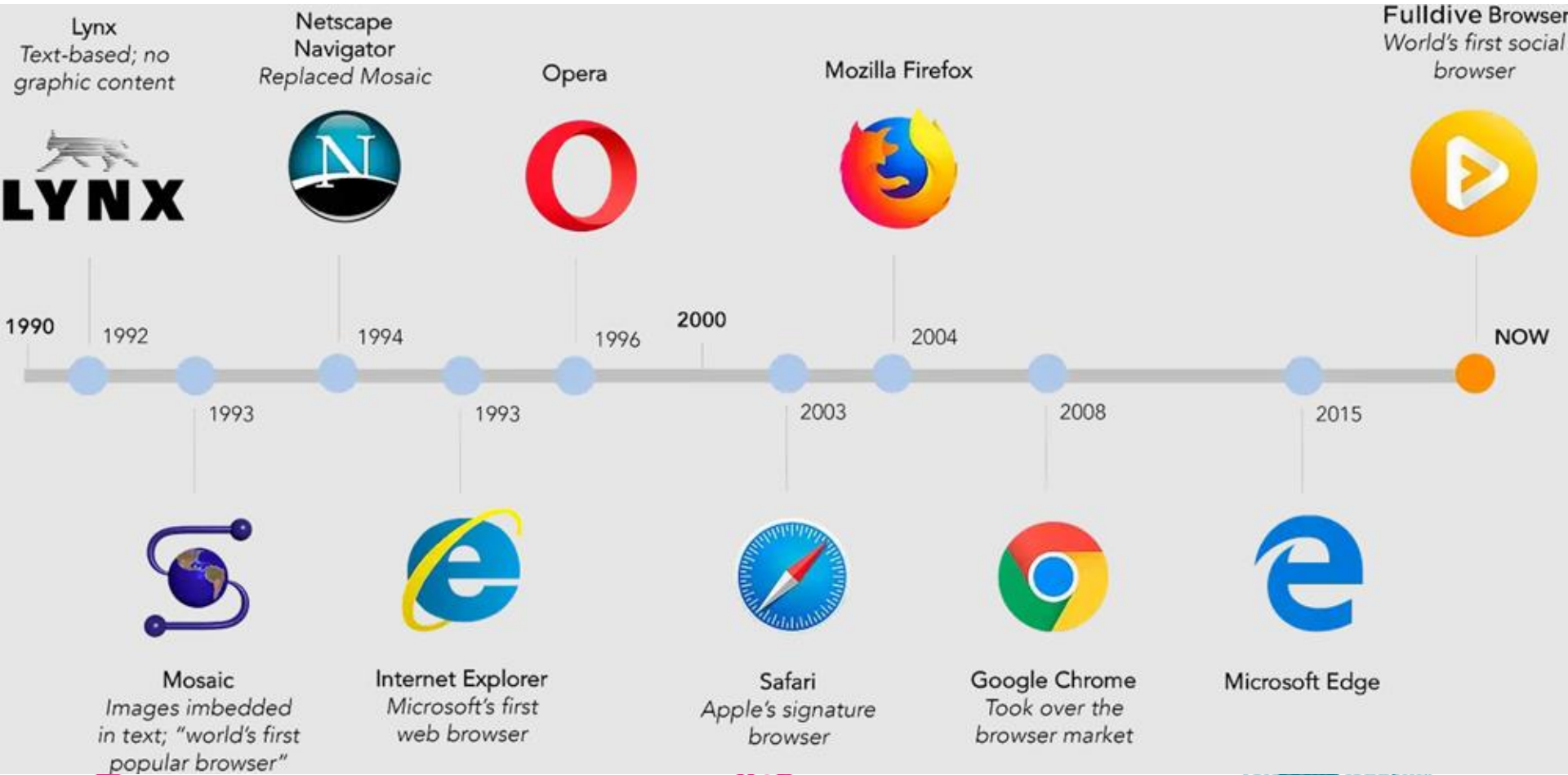
Physical

- Physical structure
- Coax, Fiber, Wireless, Hubs, Repeaters



50 YEARS OF EMAIL

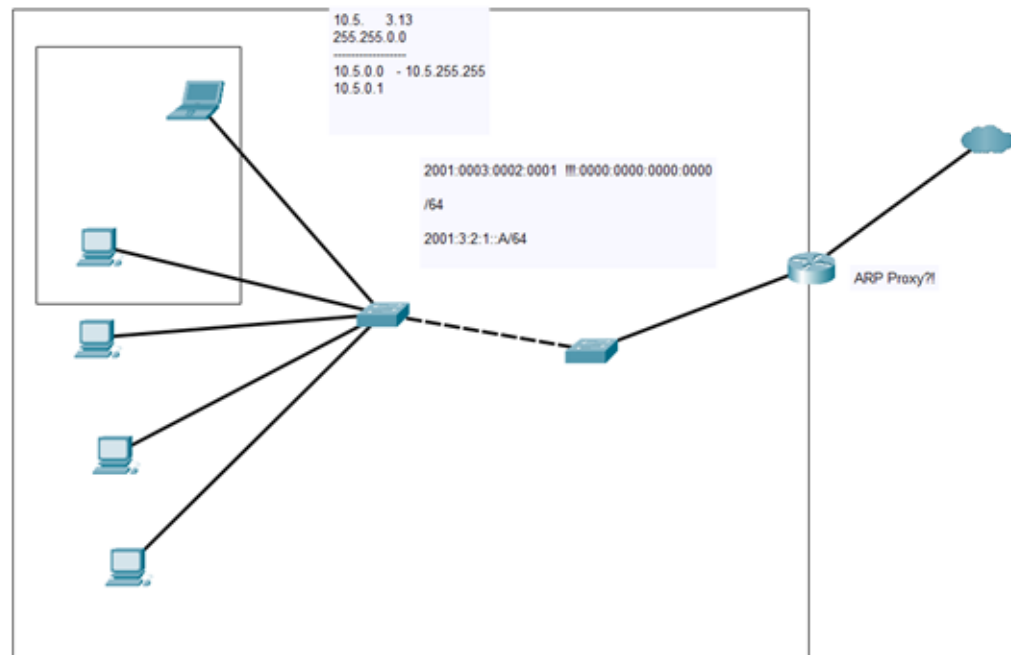




Laborator Troubleshoot

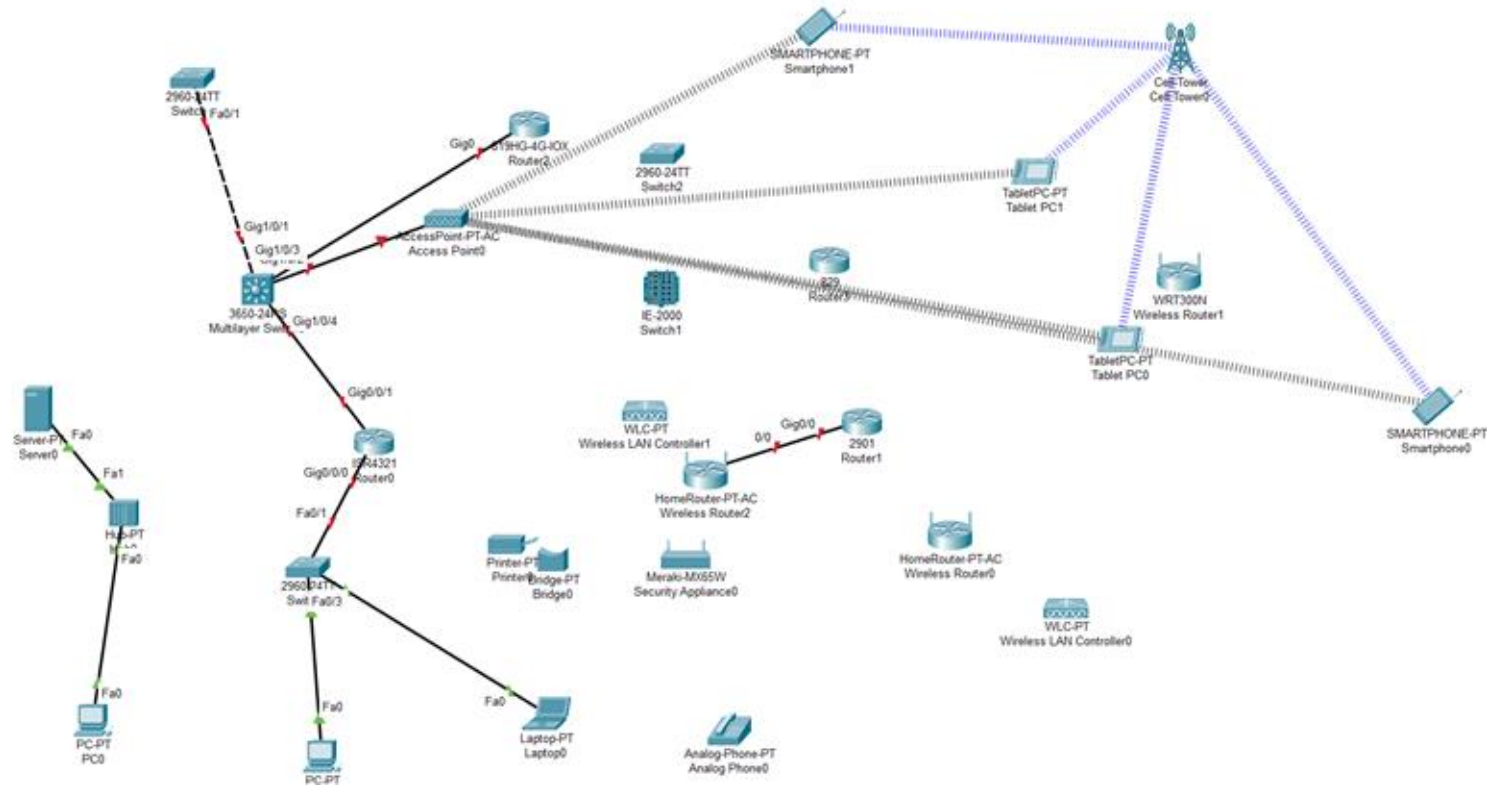
Nu merge netul?! De ce...? Putem furniza mai multe informații!

- Unelte: ping, traceroute
- Comenzi vizualizare: ipconfig, ipconfig /all
- Am greșit IP Address
- Am greșit DG
- Am greșit DNS
- Chiar și SM greșit poate duce, în absența ARP Proxy la lipsă de conexiune.
- Verificari cu IP-uri din subneturi diferite în același LAN.
- Configurare IPv6



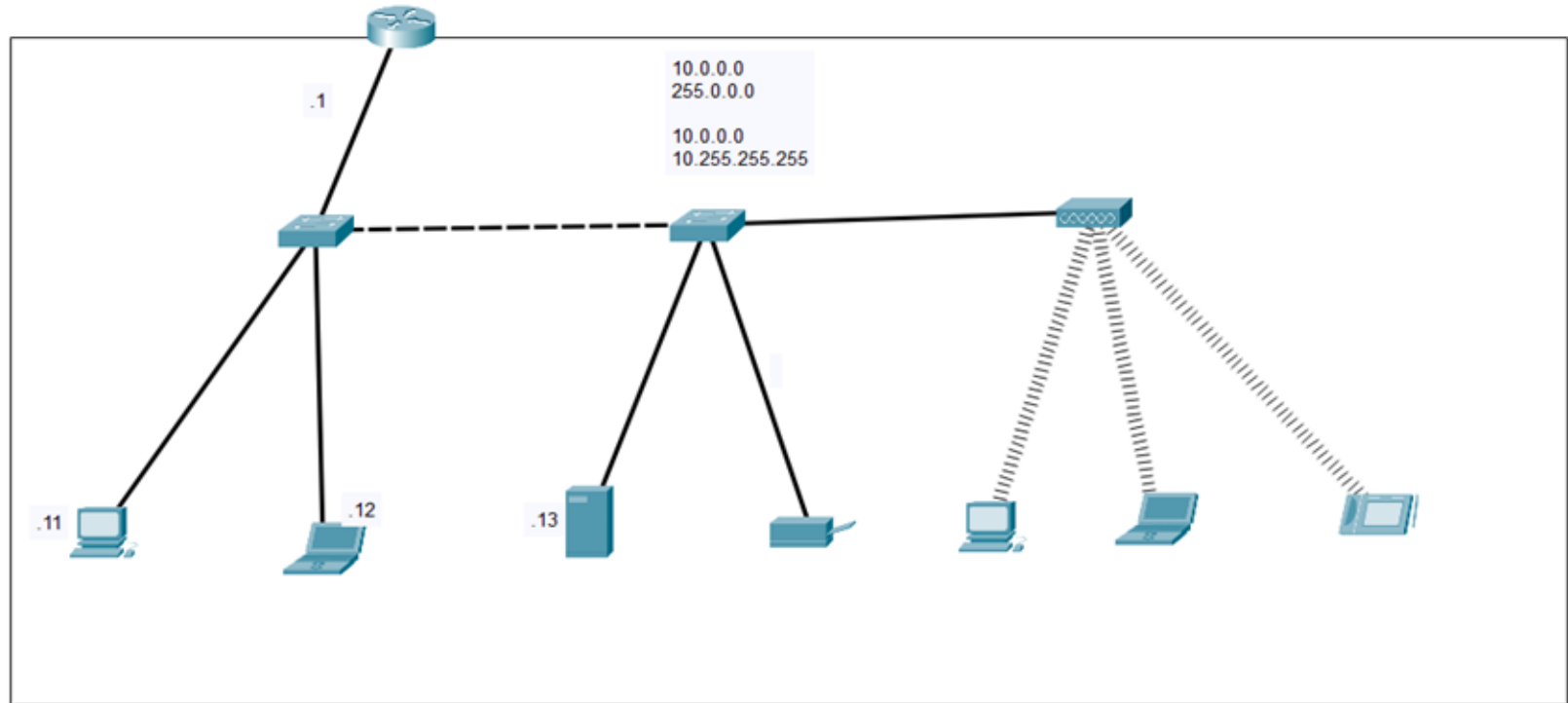
Laborator Packet Tracer

Învățăm să folosim Packet Tracerul!



Tema curs 1 Packet Tracer

Trimiteți tema până la ședința 3



Router Access Point

Adresare IP

10 . 5 . 1 .239
255.255. 0 . 0

IP uri in total : 10.5.0.0 – 10.5.255.255

IP uri folosibile: 10.5.0.1 – 10.5.255.254



Adresare IP

192.168. 0 .100

255.255. 255 . 0

IP uri in total : 192.168.0.0 – 192.168.0.255

IP uri folosibile: 192.168.0.1 – 192.168.0.254



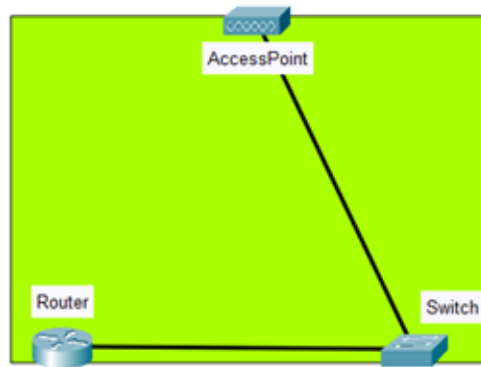
SOHO Router AP

Router+Switch+Access Point

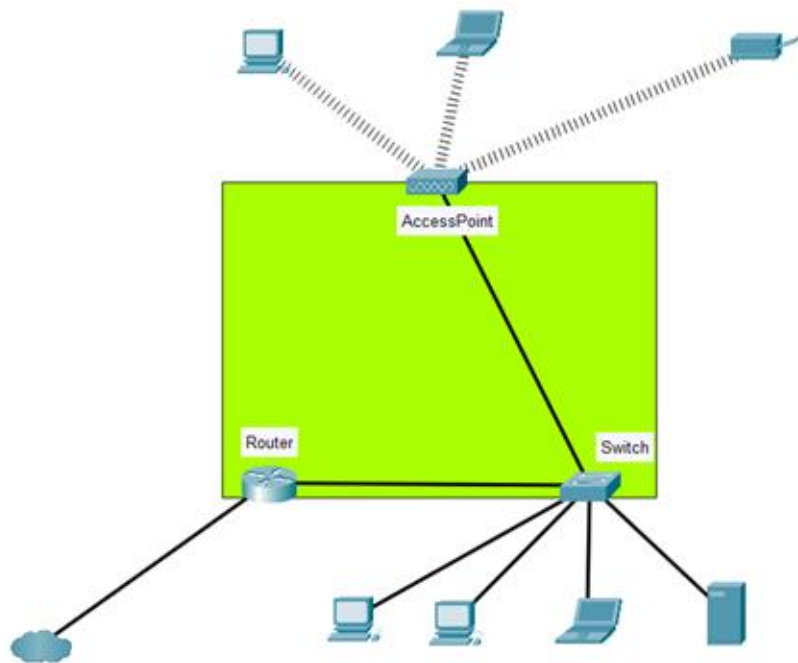


SOHO Router AP

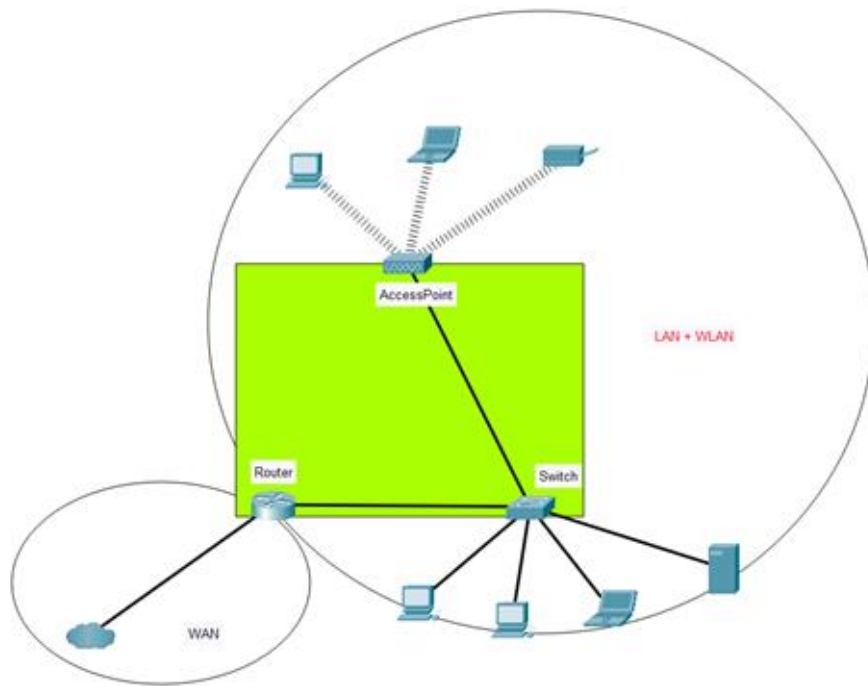
Router+Switch+Access Point



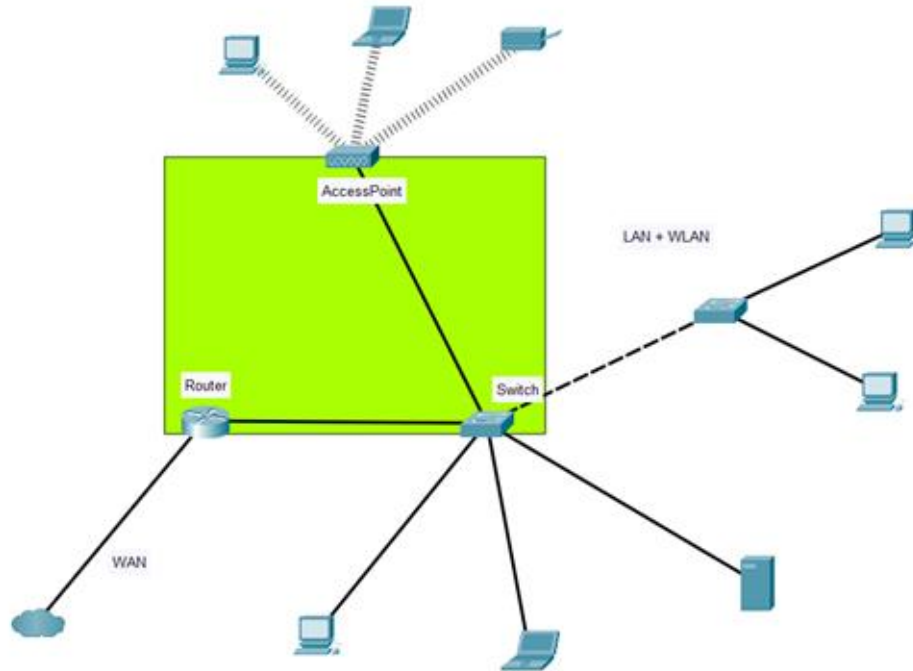
Routerul Separa Retelele si Oferă Servicii



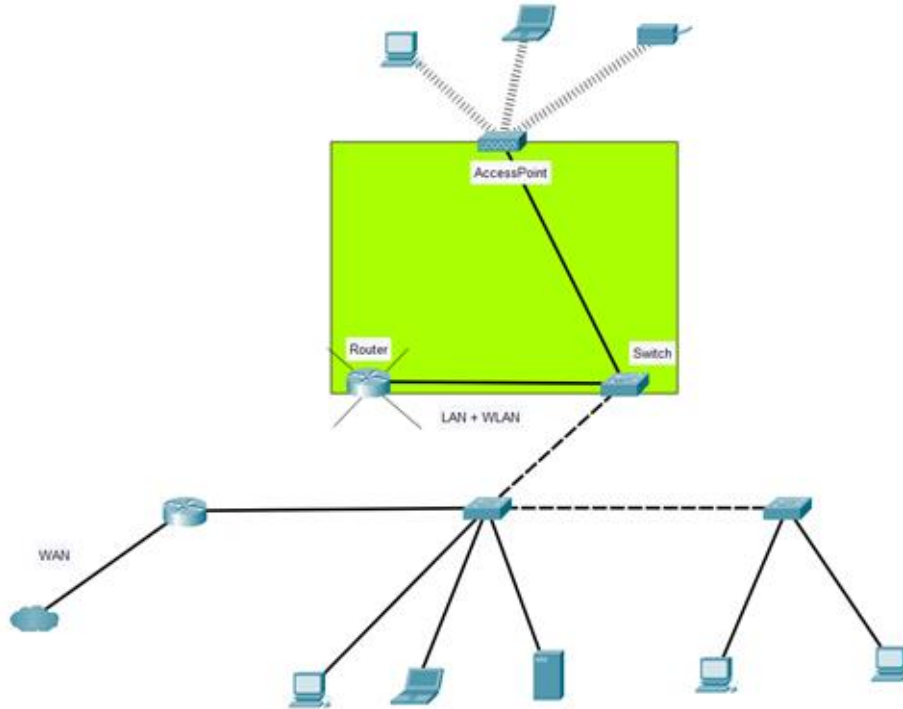
Routerul Separa Retelele si Oferă Servicii



Scenariul 1. Home Office



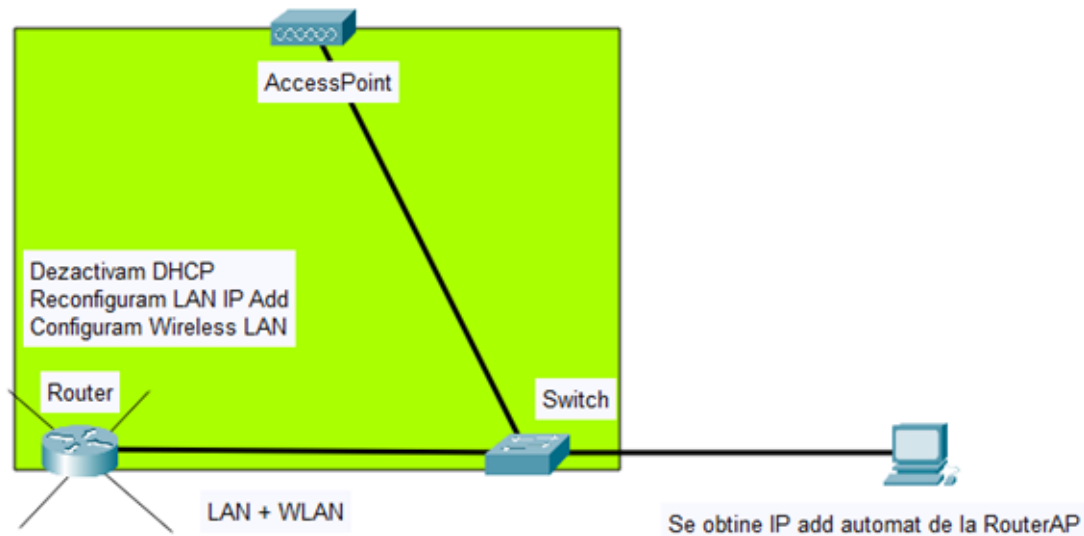
Scenariul 2. Small Office



Laborator Configurare Scenariul 2

Access Point

Configurare Scenariu AP



Transformări Binar-Zecimal-Hexazecimal

Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

1	1	1	0	0	0	1	1
—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

1	1	1	0	0	0	1	1
—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1

227



Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

0 1 0 1 0 1 1 0

— — — — — — — —
 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

128 64 32 16 8 4 2 1



Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

0 1 0 1 0 1 1 0

— — — — — — — —
 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

128 64 32 16 8 4 2 1

86



Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

1	0	1	0	0	1	0	1
—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

1. Binar > Zecimal

1 0 1 0 0 1 0 1

— — — — — — — —
 2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

128 64 32 16 8 4 2 1

165



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal

$$201 - 128 = 73$$

1

—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal

$$201-128 = 73-64=9$$

1							1
_____	_____	_____	_____	_____	_____	_____	_____
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal

$$201 - 128 = 73 - 64 = 9$$

1 1 0 0

—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal

$$201-128 = 73-64=9-8 =1-1$$

1	1	0	0	1	0	0	1
—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal

$$150 = 128 + 16 + 4 + 2$$

—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal-Hexazecimal

2. Binar < Zecimal

$$150 = 128 + 16 + 4 + 2$$

1	0	0	1	0	1	1	0
—	—	—	—	—	—	—	—
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1



Transformări Binar-Zecimal

3. Binar > Hexazecimal

0101.1011₂

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

3. Binar > Hexazecimal

0101.1011₂

5

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

3. Binar > Hexazecimal

0101.1011₂
5 B₁₆

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

3. Binar > Hexazecimal

0101.1011₂

5 B₁₆

4. Binar < Hexazecimal

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

3. Binar > Hexazecimal

0101.1011₂
5 B₁₆

4. Binar < Hexazecimal

A 6₁₆

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

3. Binar > Hexazecimal

0101.1011₂
5 B₁₆

4. Binar < Hexazecimal

A 6₁₆
1010.

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

3. Binar > Hexazecimal

0101.1011₂
5 B₁₆

4. Binar < Hexazecimal

A 6₁₆
1010.0110₂

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

5. Zecimal > Binar > Hexazecimal

100

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

5. Zecimal > Binar > Hexazecimal

$$100 = 64 + 32 + 4$$

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

5. Zecimal > Binar > Hexazecimal

$$100 = 64 + 32 + 4$$

0110.0100_2

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

5. Zecimal > Binar > Hexazecimal

$$100 = 64 + 32 + 4$$

0110.0100₂

6 4 16

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

5. Zecimal > Binar > Hexazecimal

$$100 = 64 + 32 + 4$$

0110.0100₂

6 4 16

6. Zecimal < Binar < Hexazecimal

C 4 16

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

5. Zecimal > Binar > Hexazecimal

$$100 = 64 + 32 + 4$$

0110.0100₂

6 4 16

6. Zecimal < Binar < Hexazecimal

C 4 16

1100.0100₂

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



Transformări Binar-Zecimal

5. Zecimal > Binar > Hexazecimal

$$100 = 64 + 32 + 4$$

0110.0100₂

6 4 16

6. Zecimal < Binar < Hexazecimal

C 4 16

1100.0100₂

$$128 + 64 + 4_2$$

Deci	Hexa	Binar
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111



